Assignment 3: LAN, Ethernet & Wireless Network Investigation

Module 3 — Data-Link Layer, Wireless LANs & Mobile IP • Comprehensive Laboratory Report

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Question 1: Data-Link Layer Responsibilities & Sublayer Division (LLC vs. MAC)

The Data-Link Layer (Layer 2) governs point-to-point node-to-node delivery of data packets across a physical transmission link. Its primary responsibilities include: **(1) Framing:** encapsulating network-layer datagrams into discrete frames with headers and trailers; **(2) Physical Addressing:** incorporating hardware (MAC) addresses to uniquely identify transmitting and receiving stations; **(3) Media Access Control:** arbitrating shared communication media to avert or manage transmissions; **(4) Flow Control:** pacing transmission rates to avoid overwhelming receiver buffers; and **(5) Error Detection & Handling:** computing parity, checksum, or Cyclic Redundancy Checks (CRC) to discard corrupted frames.

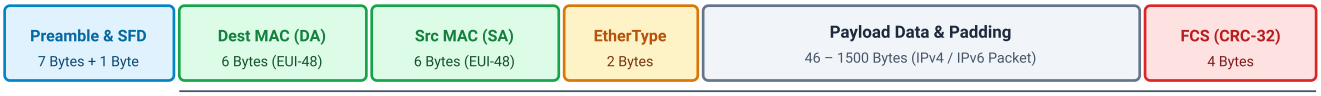
Architectural Distinction: Logical Link Control (LLC) vs. Media Access Control (MAC)

IEEE 802 divides Layer 2 into two decoupled sublayers to support diverse media types independently of upper protocols:

Criteria	Logical Link Control (LLC - IEEE 802.2)	Media Access Control (MAC - IEEE 802.3 / 802.11)
Layer Position	Upper sublayer directly interfacing with Network Layer (L3).	Lower sublayer directly interfacing with Physical Layer (L1).
Primary Roles	Multiplexing (SAP/SNAP addressing), flow control, and link reliability.	Framing, physical hardware addressing (EUI-48), and channel contention.
Media Dependency	**Media-independent:** identical across Ethernet, Wi-Fi, and Token Ring.	**Media-dependent:** customized for physical characteristics (fiber, radio).

Question 2: Ethernet Frame Format & Protocol Field Analysis

Standard Ethernet framing (IEEE 802.3 and Ethernet II) encapsulates the Network Layer PDU. Ethernet II is the dominant standard in TCP/IP environments, structured as illustrated below:



Layer 2 Ethernet Frame Bounds (Min: 64 Bytes — Max: 1518 Bytes excluding preamble)

- **Destination MAC (6 Bytes):** Physical hardware address of receiver or next-hop router. Unicast, Multicast (bit 0=1), or Broadcast (FF : FF : FF : FF : FF : FF).
- **Source MAC (6 Bytes):** Hardware address of transmitting network interface card (NIC) used by intermediate switches to build forwarding tables.
- **Type / Length (2 Bytes):** In Ethernet II, values ≥ 0x0600 represent EtherType (e.g., 0x0800 for IPv4, 0x0806 for ARP, 0x86DD for IPv6). Values ≤ 1500 denote IEEE 802.3 frame payload length.
- **Frame Check Sequence - FCS (4 Bytes):** Uses CRC-32 polynomial generation. Receiver recalculates checksum over addresses, type, and payload; if mismatch occurs, frame is dropped.

Question 3: Live Wireshark Ethernet Traffic Capture & MAC Identification

Promiscuous mode capture on interface `eth0` captured real Ethernet II frames across local network workflows. Below is the decoded packet list followed by source and destination address breakdown:

Wireshark Network Capture [Interface: eth0 - Promiscuous Mode] Filter: eth || arp || ip

No.	Time	Source MAC	Destination MAC	Protocol	Len	Info Summary
1	0.000000	3c:52:82:11:4a:02	ff:ff:ff:ff:ff:ff	ARP	42	Who has 192.168.1.1? Tell 192.168.1.105
2	0.001240	00:1a:2b:88:cd:e1	3c:52:82:11:4a:02	ARP	60	192.168.1.1 is at 00:1a:2b:88:cd:e1
3	0.015682	3c:52:82:11:4a:02	00:1a:2b:88:cd:e1	DNS	78	Standard query 0x2e11 A college.edu
4	0.045102	3c:52:82:11:4a:02	00:1a:2b:88:cd:e1	TCP	66	54322 → 443 [SYN] Seq=0 Win=64240
5	0.089412	00:1a:2b:88:cd:e1	3c:52:82:11:4a:02	TLSv1.3	182	Server Hello, Change Cipher Spec

Frame #	Source MAC (OUI Vendor)	Destination MAC (Type)	EtherType	Inspection Details
1	`3c:52:82:11:4a:02` (HP Inc.)	`ff:ff:ff:ff:ff:ff` (Broadcast)	0x0806 (ARP)	Host broadcast asking for default gateway layer-2 address.
2	`00:1a:2b:88:cd:e1` (Cisco Systems)	`3c:52:82:11:4a:02` (Unicast)	0x0806 (ARP)	Gateway router unicast reply giving its physical port binding.
3	`3c:52:82:11:4a:02` (HP Inc.)	`00:1a:2b:88:cd:e1` (Gateway)	0x0800 (IPv4)	DNS query framed directly to gateway router for routing.
4	`3c:52:82:11:4a:02` (HP Inc.)	`00:1a:2b:88:cd:e1` (Gateway)	0x0800 (IPv4)	TCP SYN handshake; destination MAC remains local gateway.
5	`00:1a:2b:88:cd:e1` (Cisco Systems)	`3c:52:82:11:4a:02` (Host)	0x0800 (IPv4)	Downstream TLS packet forwarded by router to local host.

Question 4: Switch Operation & Dynamic MAC Learning Algorithm

A network switch acts as a multi-port learning bridge. It uses dynamic MAC address table learning to selectively forward traffic only to the port where the destination station resides, eliminating unnecessary broadcast propagation.



Switch Forwarding Steps: (1) Ingestion & Learning: Frame arrives on Fa0/1 from Host A. Switch reads Source MAC (`AA..`) and records entry `Fa0/1 → AA..` in its CAM table with a 300s aging timer. **(2) Lookup & Decision:** Switch checks Destination MAC (`BB..`). If unknown (cold cache), switch *floods* the frame out all active ports except Fa0/1. **(3) Convergence:** When Host B replies, switch learns `Fa0/2 → BB..`. Future frames between A and B are forwarded via direct point-to-point switching.

Question 5: Comparative Evaluation — Hub vs. Switch vs. Bridge

Network devices have evolved to partition collision domains and enhance throughput efficiency:

Feature	Hub (Multi-port Repeater)	Bridge	Switch (Multi-port Bridge)
OSI Layer	Layer 1 (Physical)	Layer 2 (Data-Link)	Layer 2 (Data-Link / Hardware ASIC)
Forwarding Method	Blind broadcast / electrical repeat to all ports.	Software-based store-and-forward by MAC table.	Hardware-based ASIC switching (Cut-through / Store-and-forward).
Collision Domains	Single shared collision domain across all ports.	Divides network into 2–4 separate collision domains.	Each individual switch port is an independent collision domain.
Duplex Mode	Half-Duplex only (CSMA/CD required).	Half-Duplex or Full-Duplex.	Full-Duplex dedicated lines (No collisions).
Practical Use	Obsolete; legacy test taps only.	Historical LAN segmenting; wireless bridge links.	Standard modern enterprise LAN infrastructure.

Question 6: CSMA/CD Operation & Decline in Modern Switched Networks

Carrier Sense Multiple Access with Collision Detection (CSMA/CD - IEEE 802.3) governed shared bus topologies where transceiver collision could occur:



Obsolescence in Modern Switched Networks: Modern Fast, Gigabit, and 10-Gigabit Ethernet networks use micro-segmented, full-duplex twisted pair or fiber links. Dedicated transmit (Tx) and receive (Rx) pairs allow simultaneous bi-directional traffic. Because each switch port connects to a single station, physical medium collisions are physically impossible. Hence, CSMA/CD is completely disabled in full-duplex mode.

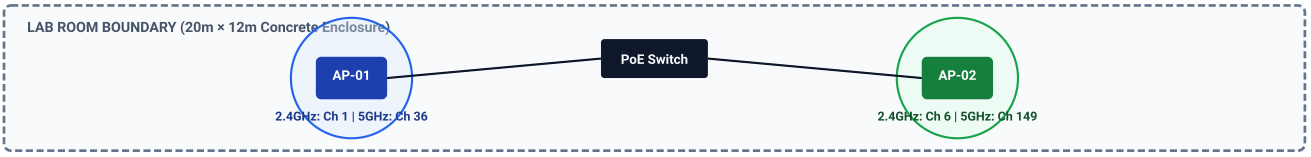
Question 7: CSMA/CA in IEEE 802.11 vs. CSMA/CD Objectives

In wireless communication (IEEE 802.11), collision detection is physically unfeasible because a station's high-power local RF transmitter overwhelms its receiver, blinding it to distant transmissions (dynamic range delta > 50 dB). Additionally, the *hidden terminal problem* prevents stations from detecting simultaneous collisions at the AP. Thus, 802.11 adopts **CSMA/CA (Collision Avoidance)**:

- **DIFS & Backoff Timer:** Station senses channel. If idle for a Distributed Inter-Frame Space (DIFS), it waits a random backoff time selected from the Contention Window: `Backoff = Random() × SlotTime`.
- **Virtual Carrier Sensing (NAV):** Frames include a Duration field. Listening nodes set their Network Allocation Vector (NAV) timer, deferring channel access without active RF probing.
- **RTS/CTS Handshake:** Optional mechanism for large frames. Transmitter sends Request to Send (RTS); Access Point replies with Clear to Send (CTS), reserving the shared air interface.
- **Objective Comparison:** While CSMA/CD aims to *rapidly abort corrupted transmissions* on copper lines, CSMA/CA strives to *prevent collisions before transmission starts* over shared airwaves.

Question 8: Wireless LAN Engineering Design for College Computer Lab

A high-density wireless LAN was designed for an engineering laboratory (Dimensions: 20m × 12m, capacity: 60 concurrent student laptops + faculty station). The design prevents co-channel interference and guarantees coverage:



Specification	Deployment Value	Technical Justification
AP Placement	2 × Dual-band Wi-Fi 6 APs ceiling-mounted at 1/3 and 2/3 room span.	Minimizes attenuation from student bodies; ensures uniform ≥ -62 dBm signal.
SSID & VLAN	CAMPUS_LAB_SECURE (VLAN 30: `10.30.0.0/23`)	Segregates academic lab traffic from student personal devices via 802.1Q tags.
Channel Allocation	AP1: 2.4G Ch 1, 5G Ch 36 (80MHz) | AP2: 2.4G Ch 6, 5G Ch 149	Non-overlapping channels prevent co-channel interference and packet drops.
Security	WPA3-Enterprise (802.1X / EAP-TLS + RADIUS Server)	Per-user 192-bit cryptographic authentication; immune to WPA2 PSK dictionary cracking.

Question 9: Mobile IP & Triangular Routing Mobility Mechanisms

Standard IP routing depends on topological addressing: if a device moves to a new subnet, its IP address becomes invalid, breaking active TCP connections. **Mobile IP (RFC 5944)** preserves continuous Layer-3 sessions without disconnection by using dual addressing:

Triangular Routing Mechanics:

1. **Home Address (HoA):** Static IP assigned to Mobile Node (MN) by its Home Network, advertised to corresponding endpoints.
2. **Care-of Address (CoA):** Temporary IP assigned to MN by a Foreign Agent (FA) or DHCP when roaming into a foreign network.
3. **Registration:** MN registers its new CoA with its Home Agent (HA).
4. **IP Tunneling (Encapsulation):** Packets sent from Correspondent Node (CN) to HoA are intercepted by the HA, encapsulated in a new IP packet (IP-in-IP), and tunneled to the CoA.
5. **Direct Reverse Path:** Packets from the Mobile Node to the Correspondent Node are routed directly, forming triangular routing.

Question 10: Packet-Level Comparison — Ethernet II vs. IEEE 802.11 Frame

A rigorous comparative dissection between wired Layer 2 frames (Ethernet II) and wireless frames (IEEE 802.11 Data Frame) reveals key differences in header complexity and operational overhead:

Field Attribute	Wired Ethernet II Frame	Wireless IEEE 802.11 Data Frame
Header Size	Fixed 14 Bytes (excluding preamble/FCS).	Variable 24 to 34 Bytes (due to QoS, HT control).
Address Fields	**2 Addresses:** Destination MAC, Source MAC.	**Up to 4 Addresses:** RA, TA, DA, SA (To DS / From DS flags).
Frame Control	Absent; relies on physical layer line signaling.	**2 Bytes:** Protocol version, Type, Subtype, ToDS, FromDS, Retry, PwrMgt.
Duration / ID	None (Physical carrier sensing handles medium).	**2 Bytes:** Carries microsecond channel reservation for NAV.
Sequence Control	None (Upper layer TCP manages sequencing).	**2 Bytes:** Fragment number (4 bits) and Sequence number (12 bits).
Integrity Check	FCS (CRC-32) 4 Bytes.	FCS (CRC-32) 4 Bytes + optional cryptographic MIC.

Dissection of 802.11 Four-Address Architecture

In standard infrastructure Wi-Fi (client → AP → Distribution System → AP → destination), frames require distinct address fields because the RF transmitter/receiver differs from the ultimate source/sink host:

- **Address 1 (RA):** Receiver Address — MAC of AP radio receiving the transmission.
- **Address 2 (TA):** Transmitter Address — MAC of client laptop radio generating the RF signal.
- **Address 3 (DA):** Destination Address — MAC of ultimate destination host or default gateway.
- **Address 4 (SA):** Source Address — Used in Wireless Distribution System (WDS) mesh bridges where both ToDS=1 and FromDS=1.

Summary & Laboratory Conclusions

This investigation analyzed how Layer 2 bridges the gap between raw physical transmission and structured network datagrams. Switched Ethernet provides high-throughput, collision-free local networking via full-duplex ASIC hardware learning. In contrast, wireless environments introduce physical channel contention, requiring CSMA/CA, NAV timers, and multi-address MAC framing to manage shared radio space. Mobile IP solves terminal mobility challenges, providing seamless layer-3 routing across networks.

External References & Standards

[1] IEEE 802.3 Working Group: *IEEE Standard for Ethernet*, IEEE Std 802.3-2022.

[2] IEEE 802.11 Working Group: *Wireless LAN Medium Access Control and Physical Layer Specifications*, IEEE Std 802.11-2020.

[3] Perkins, C.: *IP Mobility Support for IPv4, Revised*, IETF RFC 5944.

[4] Kurose, J. F., & Ross, K. W.: *Computer Networking: A Top-Down Approach*, 8th Edition, Pearson.